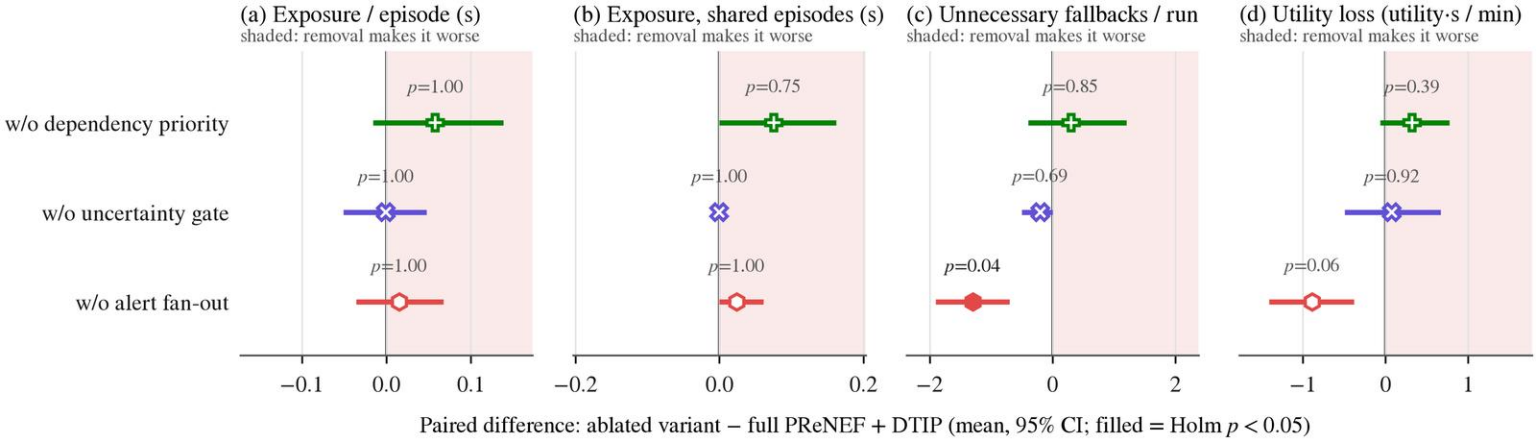
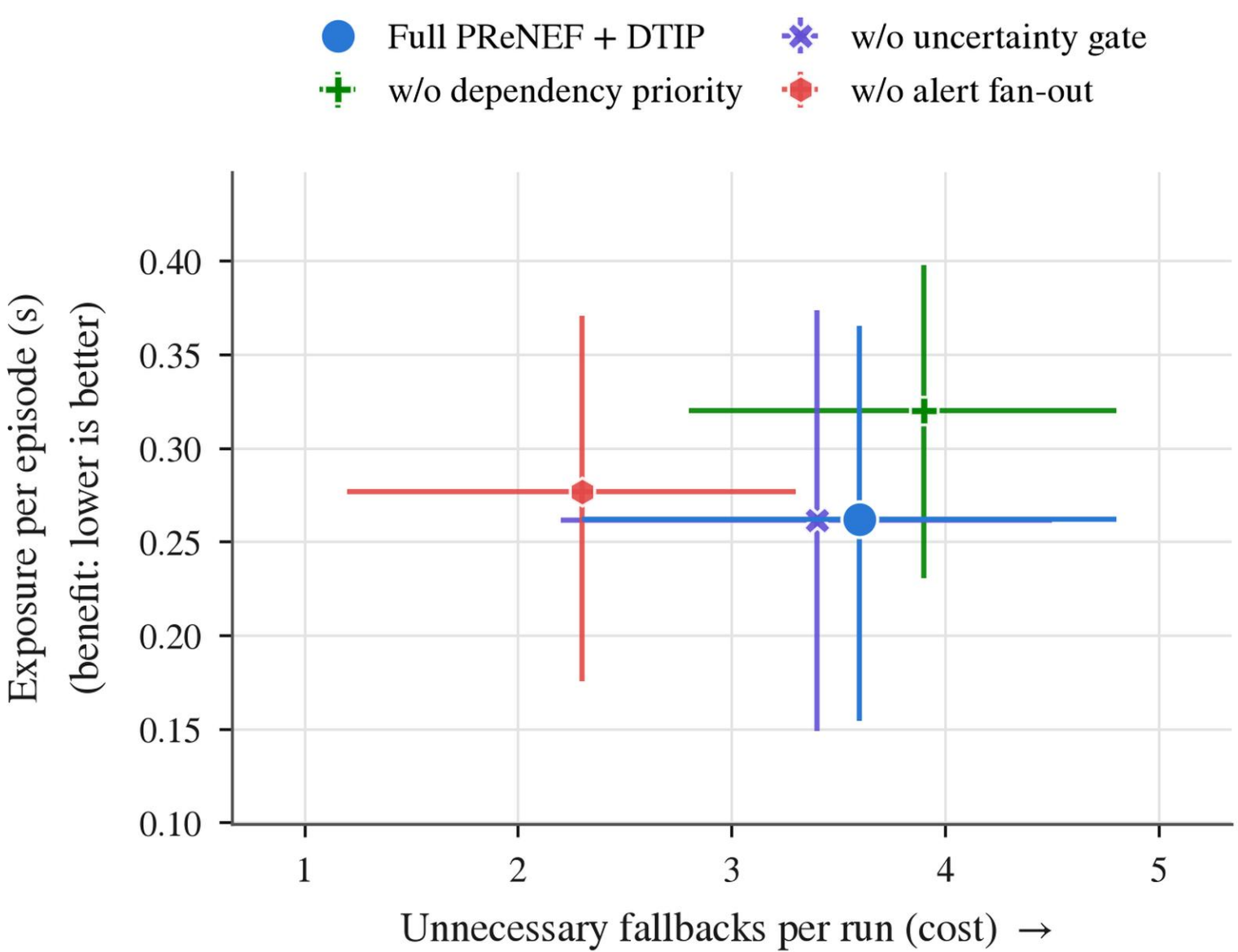
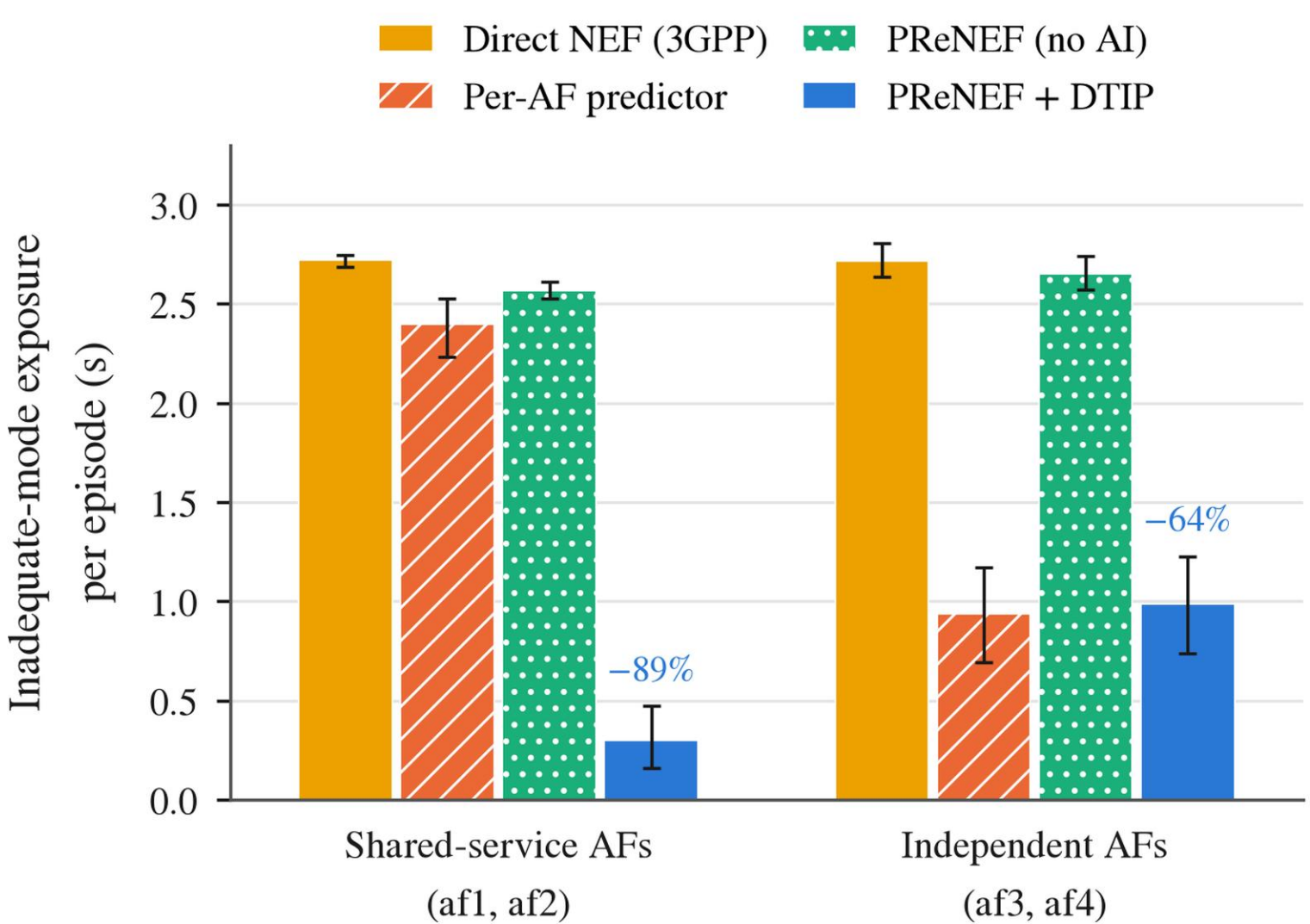
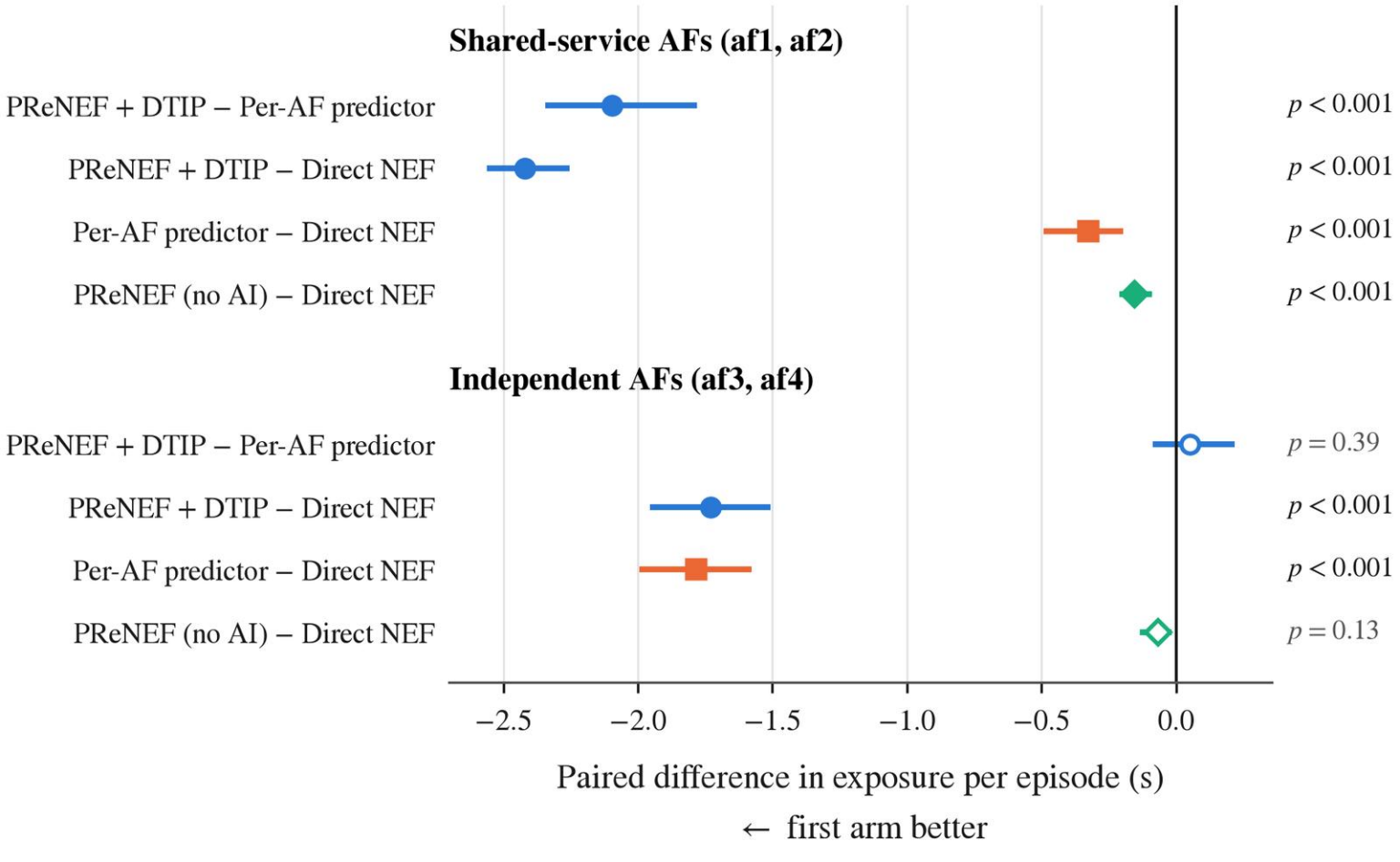
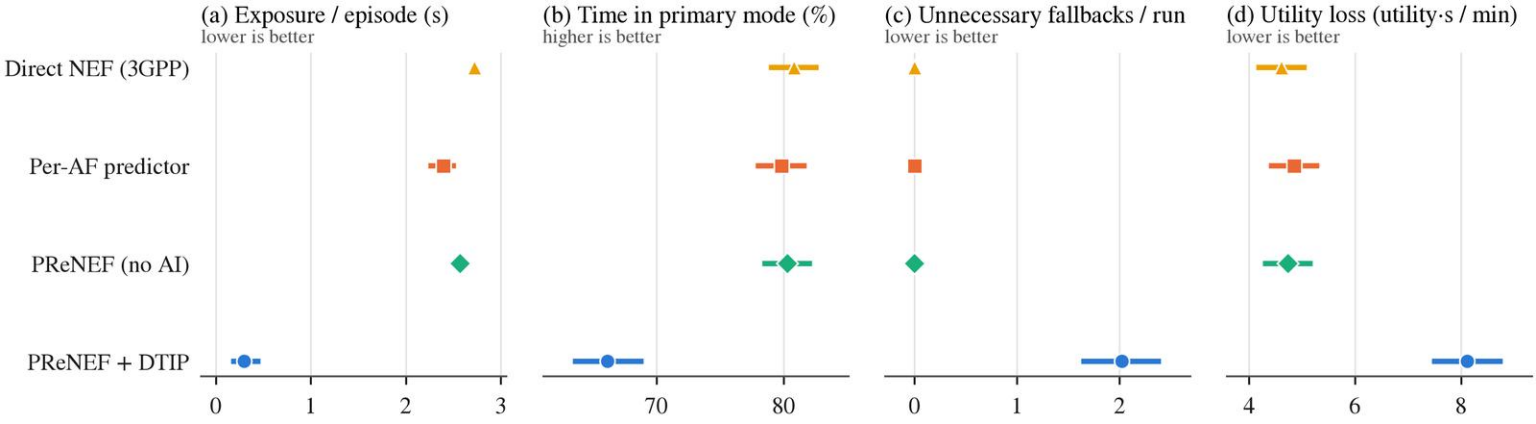
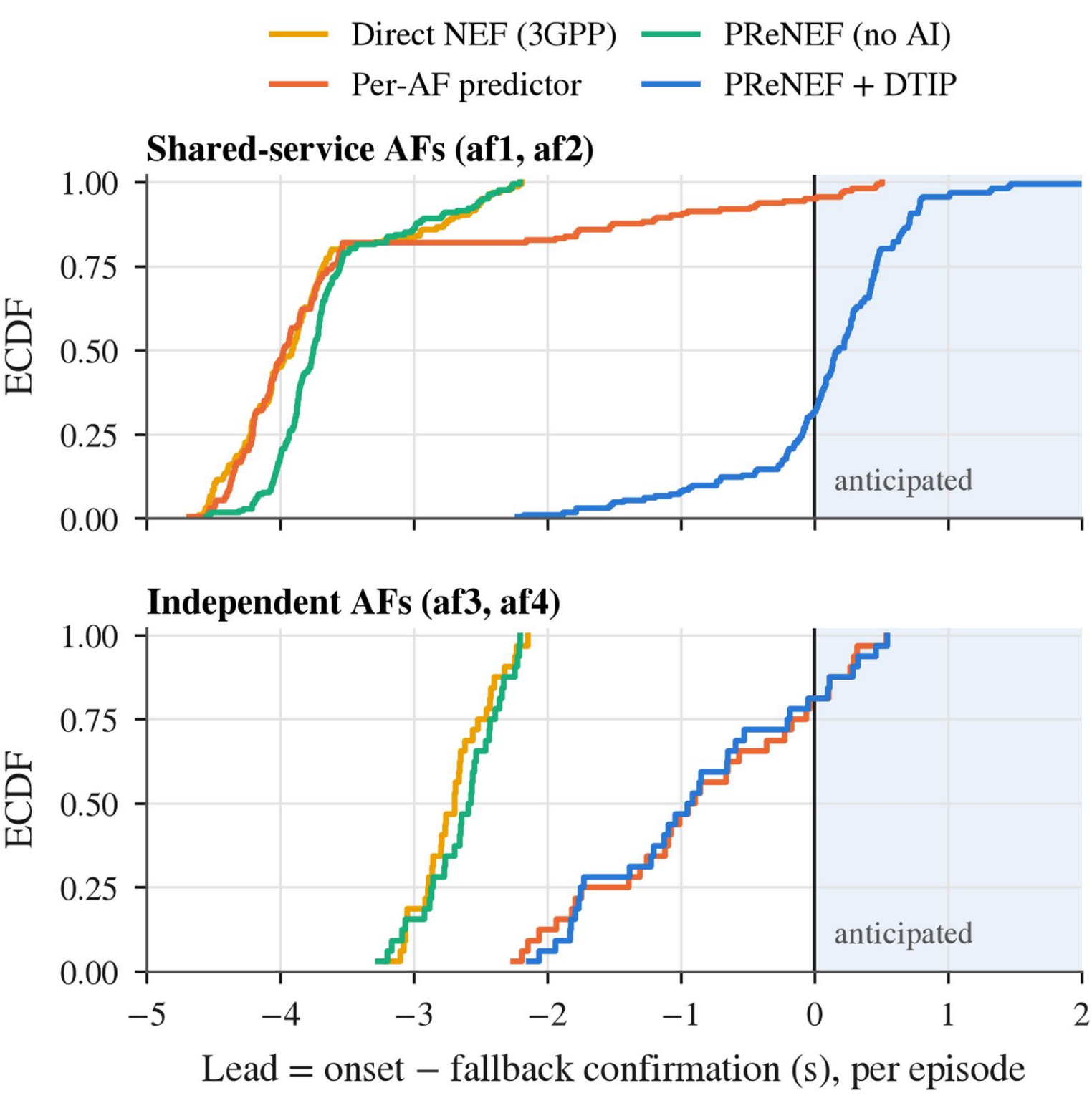

(a) Exposure / episode (s)
shaded: removal makes it worse
(b) Exposure, shared episodes (s)
shaded: removal makes it worse
(c) Unnecessary fallbacks / run
shaded: removal makes it worse
(d) Utility loss (utility·s / min)
shaded: removal makes it worse
w/o dependency priority
w/o uncertainty gate
w/o alert fan-out
p=1.00
p=1.00
p=1.00
p=0.75
p=1.00
p=1.00
p=0.85
p=0.69
p=0.04
p=0.39
p=0.92
p=0.06
−0.1
0.0
0.1
−0.2
0.0
0.2
−2
0
2
−1
0
1
Paired difference: ablated variant − full PReNEF + DTIP (mean, 95% CI; filled = Holm p < 0.05)

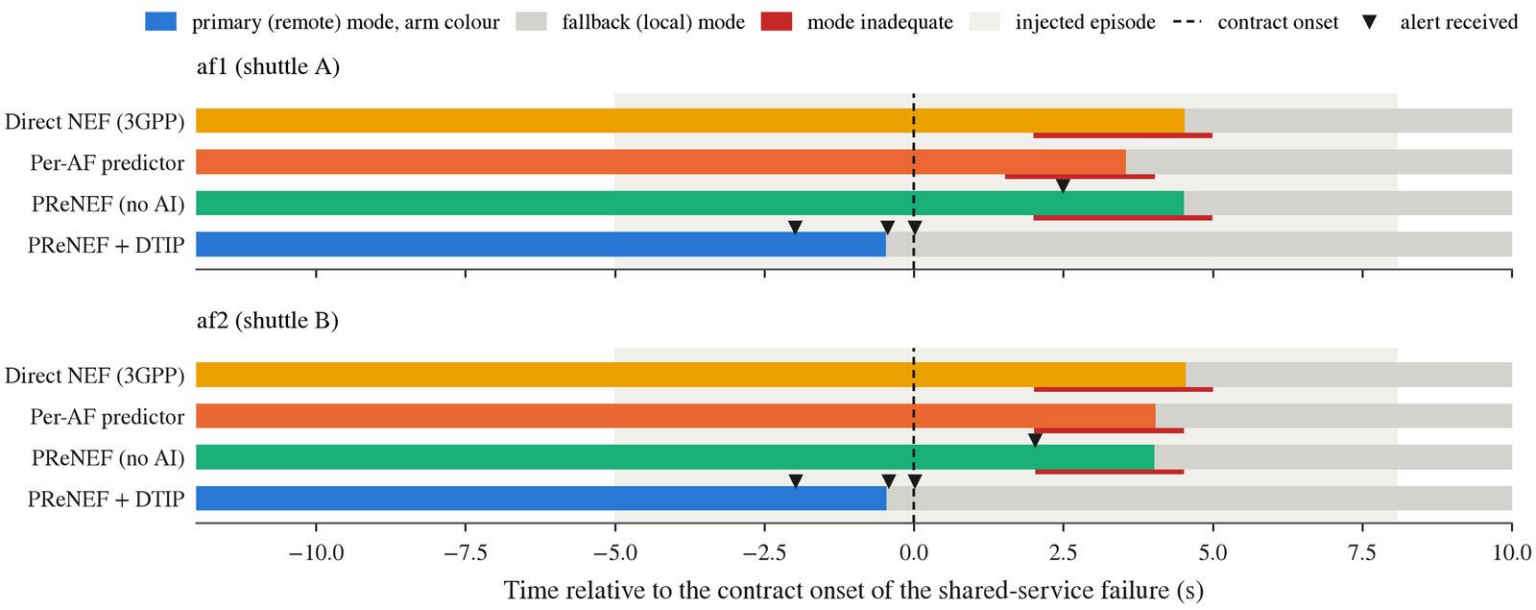

Full PReNEF + DTIP
w/o uncertainty gate
w/o dependency priority
w/o alert fan-out
Exposure per episode (s)
(benefit: lower is better)
0.40
0.35
0.30
0.25
0.20
0.15
0.10
1
2
3
4
5
Unnecessary fallbacks per run (cost) →

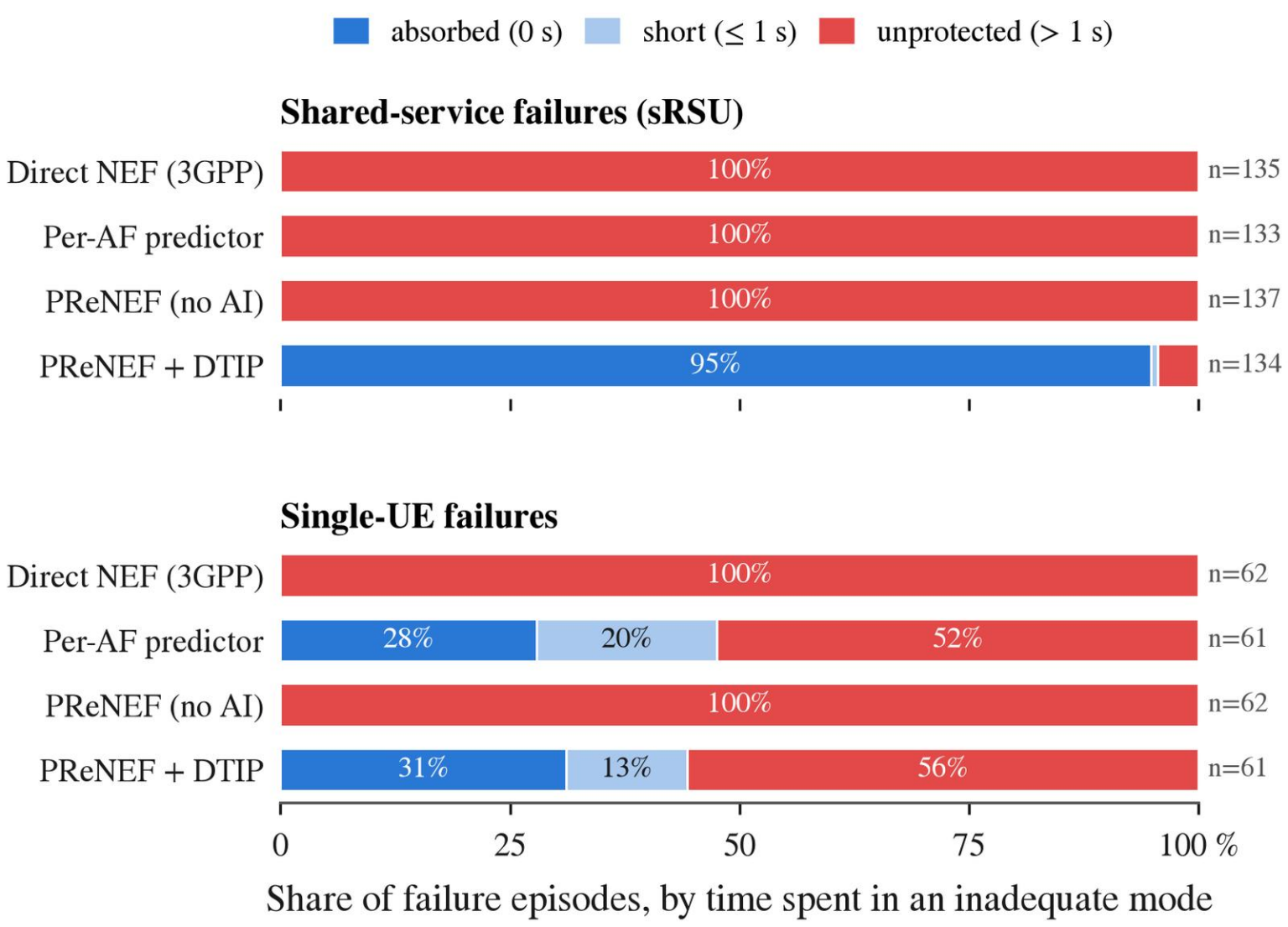

Direct NEF (3GPP)
PReNEF (no AI)
Per-AF predictor
PReNEF + DTIP
Inadequate-mode exposure per episode (s)
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−89%
−64%
Shared-service AFs
(af1, af2)
Independent AFs
(af3, af4)

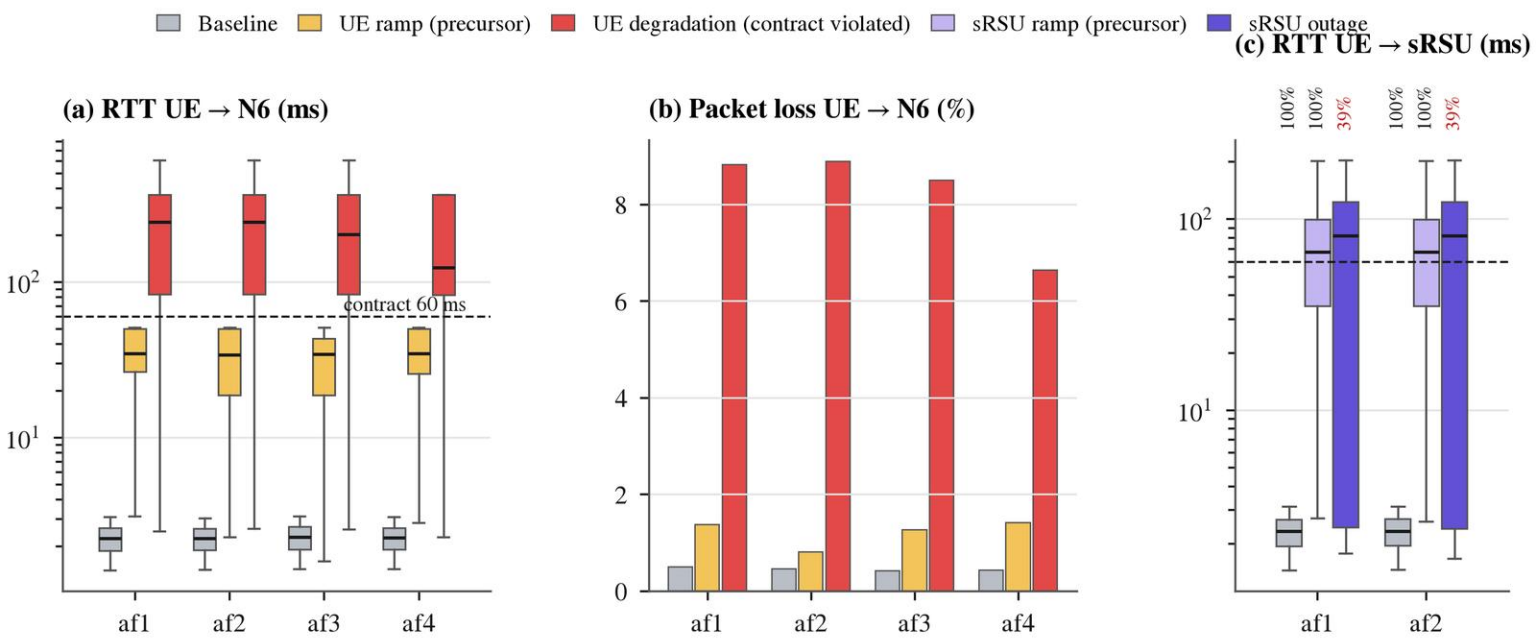

Shared-service AFs (af1, af2)
PReNEF + DTIP − Per-AF predictor
PReNEF + DTIP − Direct NEF
Per-AF predictor − Direct NEF
PReNEF (no AI) − Direct NEF
p < 0.001
p < 0.001
p < 0.001
p < 0.001
Independent AFs (af3, af4)
PReNEF + DTIP − Per-AF predictor
PReNEF + DTIP − Direct NEF
Per-AF predictor − Direct NEF
PReNEF (no AI) − Direct NEF
p = 0.39
p < 0.001
p < 0.001
p = 0.13
−2.5
−2.0
−1.5
−1.0
−0.5
0.0
Paired difference in exposure per episode (s)
← first arm better


(a) Exposure / episode (s)
lower is better
(b) Time in primary mode (%)
higher is better
(c) Unnecessary fallbacks / run
lower is better
(d) Utility loss (utility·s / min)
lower is better
Direct NEF (3GPP)
Per-AF predictor
PReNEF (no AI)
PReNEF + DTIP
0
1
2
3
70
80
0
1
2
4
6
8

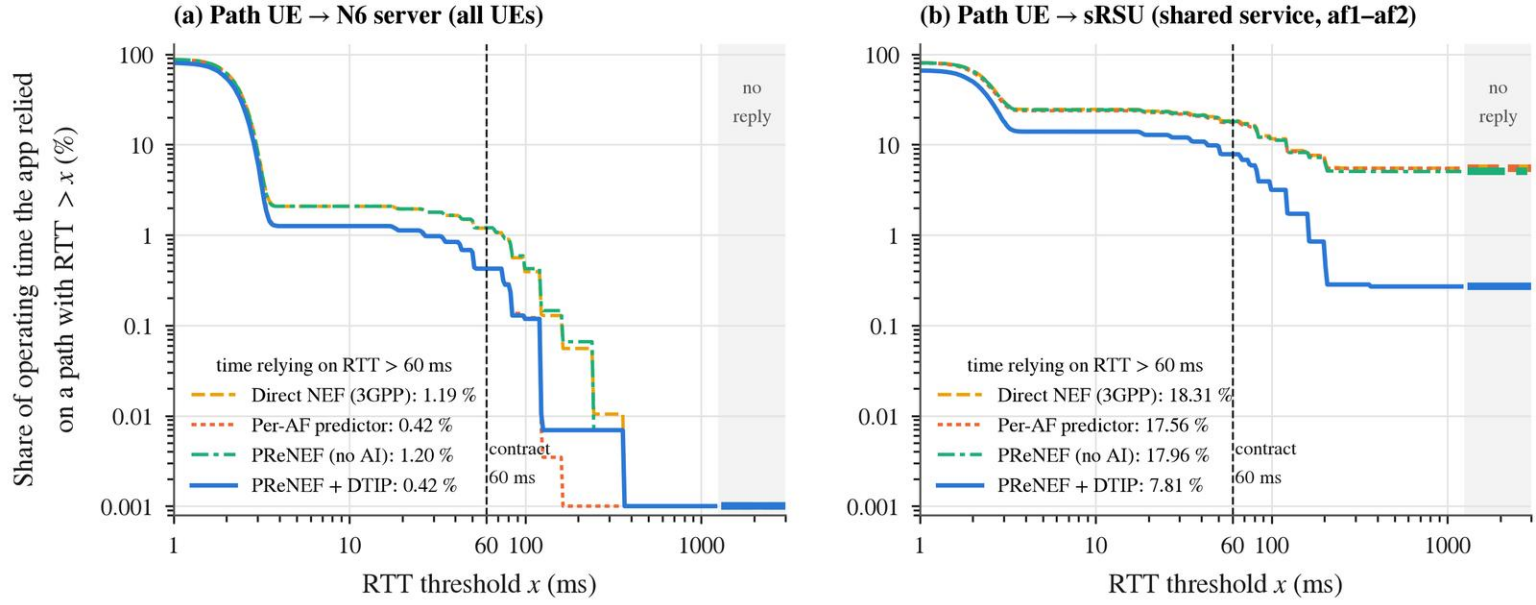

Direct NEF (3GPP)
PReNEF (no AI)
Per-AF predictor
PReNEF + DTIP
Shared-service AFs (af1, af2)
ECDF
1.00
0.75
0.50
0.25
0.00
anticipated
Independent AFs (af3, af4)
ECDF
1.00
0.75
0.50
0.25
0.00
anticipated
−5
−4
−3
−2
−1
0
1
2
Lead = onset − fallback confirmation (s), per episode

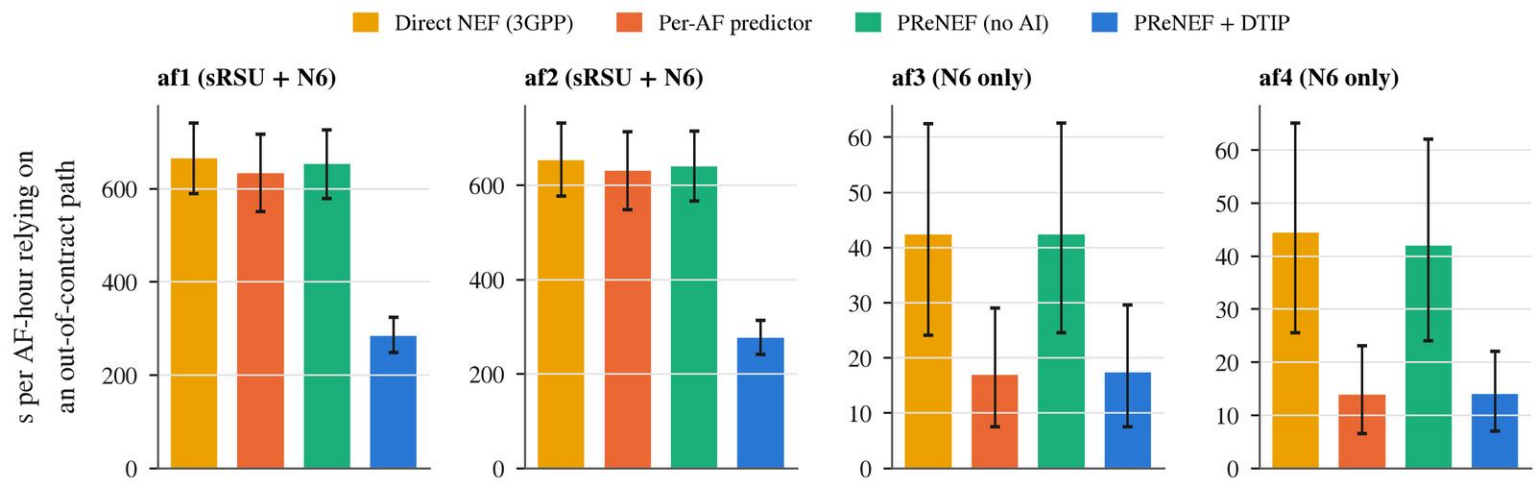

primary (remote) mode, arm colour
fallback (local) mode
mode inadequate
injected episode
contract onset
alert received
af1 (shuttle A)
Direct NEF (3GPP)
Per-AF predictor
PReNEF (no AI)
PReNEF + DTIP
af2 (shuttle B)
Direct NEF (3GPP)
Per-AF predictor
PReNEF (no AI)
PReNEF + DTIP
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
Time relative to the contract onset of the shared-service failure (s)

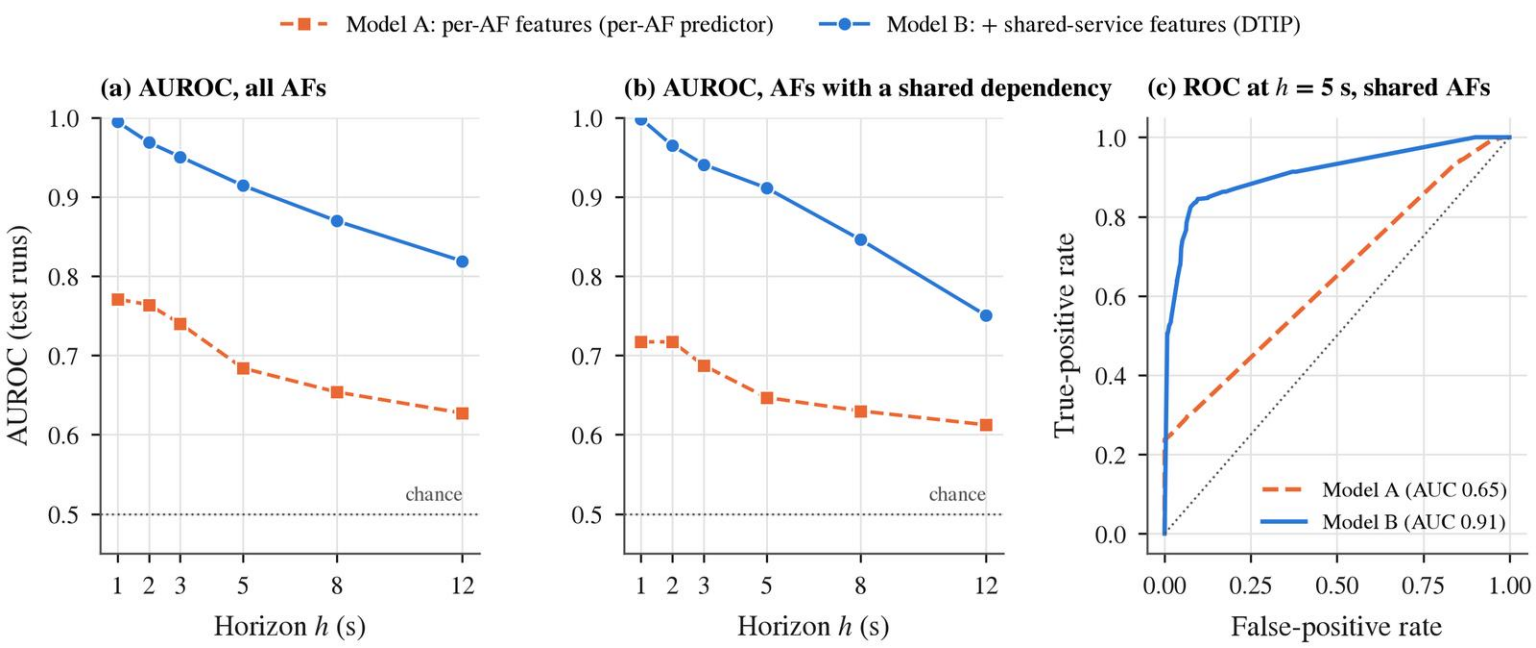

absorbed (0 s)
short (≤ 1 s)
unprotected (> 1 s)
Shared-service failures (sRSU)
Direct NEF (3GPP)
100%
n=135
Per-AF predictor
100%
n=133
PReNEF (no AI)
100%
n=137
PReNEF + DTIP
95%
n=134
Single-UE failures
Direct NEF (3GPP)
100%
n=62
Per-AF predictor
28%
20%
52%
n=61
PReNEF (no AI)
100%
n=62
PReNEF + DTIP
31%
13%
56%
n=61
0
25
50
75
100 %
Share of failure episodes, by time spent in an inadequate mode

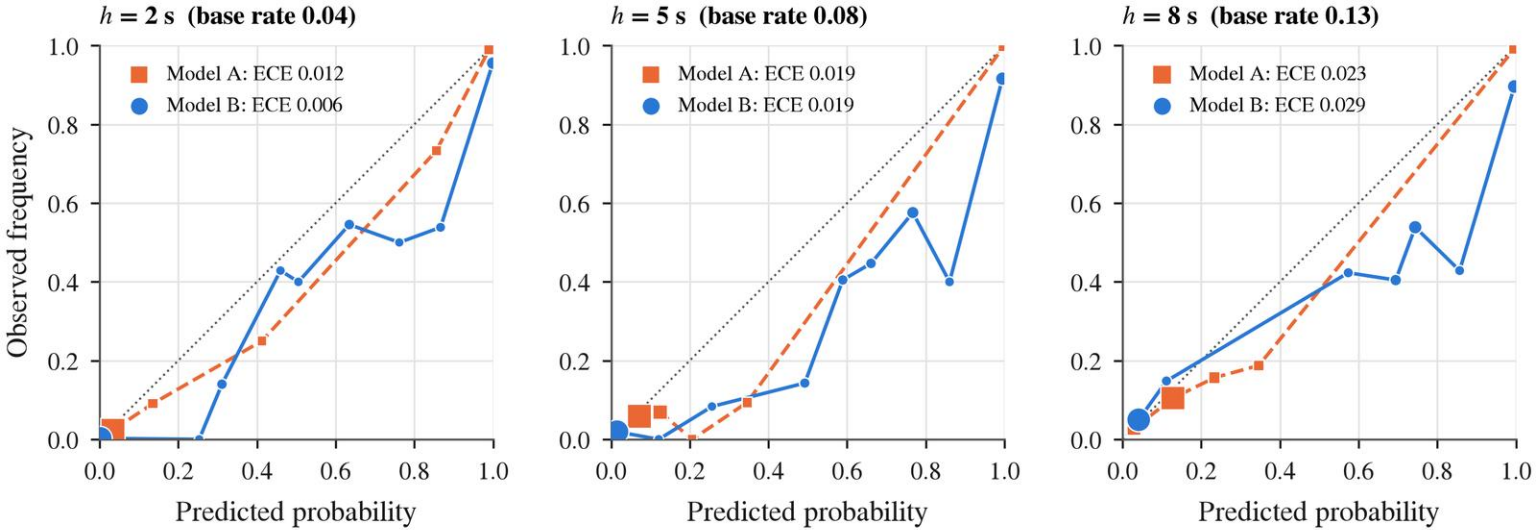


Network conditions per UE and scenario phase, pooled over arms (identical by design). Boxes: median, IQR, whiskers 5–95 %.

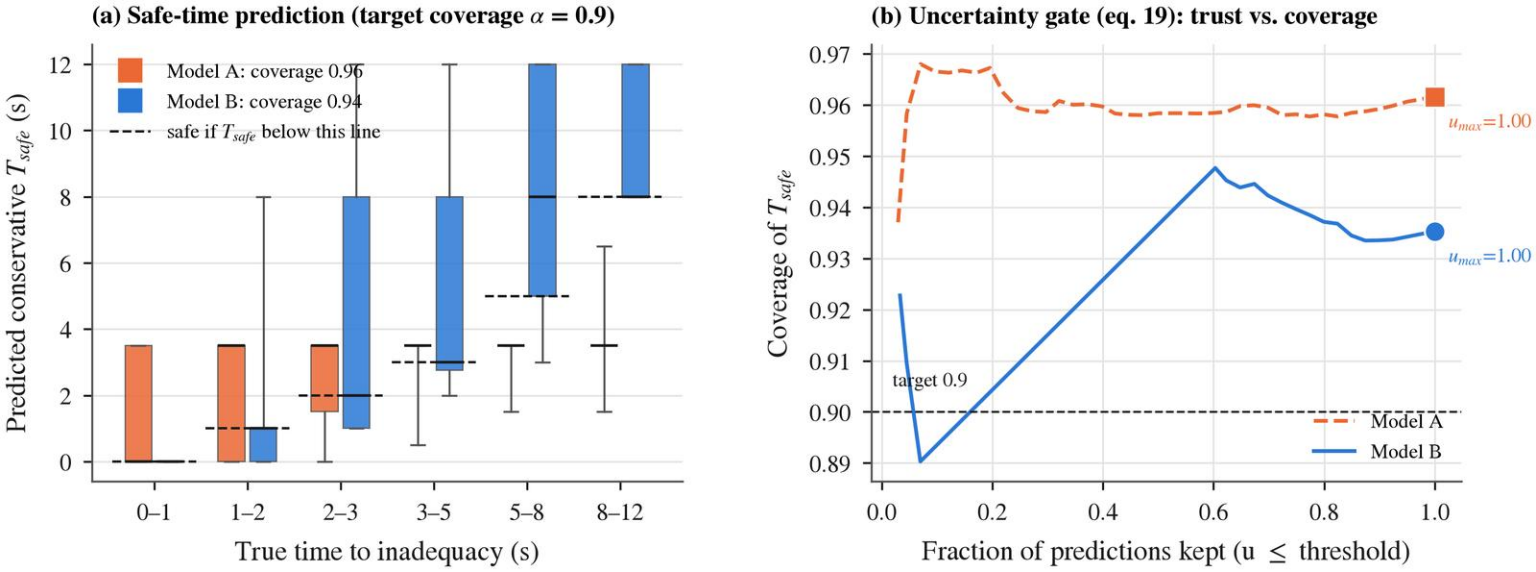

(a) Path UE → N6 server (all UEs)
(b) Path UE → sRSU (shared service, af1–af2)
Share of operating time the app relied on a path with RTT > x (%)
100
10
1
0.1
0.01
0.001
1
10
60
100
1000
RTT threshold x (ms)
no reply
time relying on RTT > 60 ms
Direct NEF (3GPP): 1.19 %
Per-AF predictor: 0.42 %
PReNEF (no AI): 1.20 %
PReNEF + DTIP: 0.42 %
contract 60 ms
Direct NEF (3GPP): 18.31 %
Per-AF predictor: 17.56 %
PReNEF (no AI): 17.96 %
PReNEF + DTIP: 7.81 %
Left end of each curve = share of time in the network-dependent (remote) mode; lower tail = fewer seconds spent depending on an out-of-contract path.

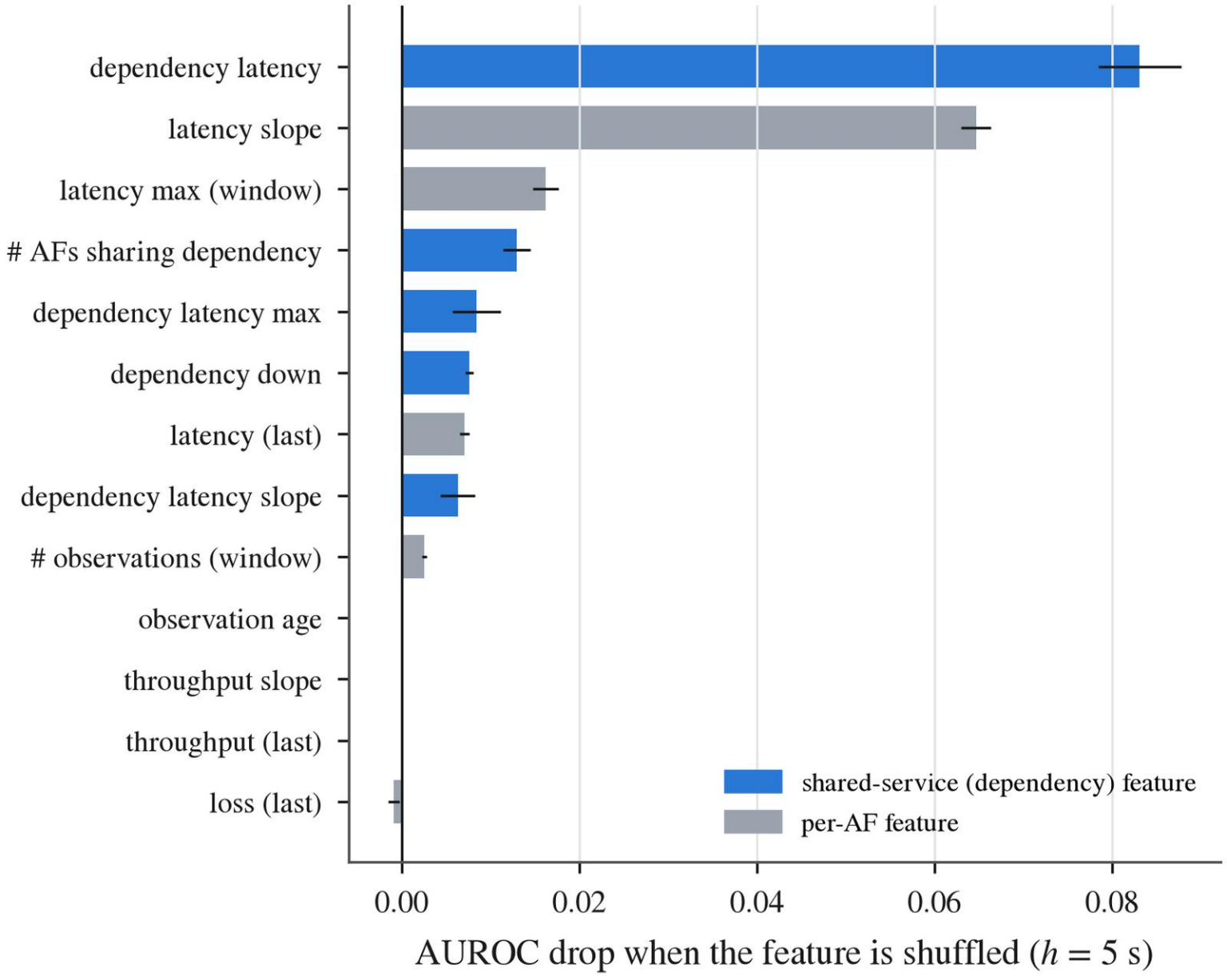

Direct NEF (3GPP)
Per-AF predictor
PReNEF (no AI)
PReNEF + DTIP
af1 (sRSU + N6)
af2 (sRSU + N6)
af3 (N6 only)
af4 (N6 only)
s per AF-hour relying on
an out-of-contract path
0
200
400
600
10
20
30
40
50
60

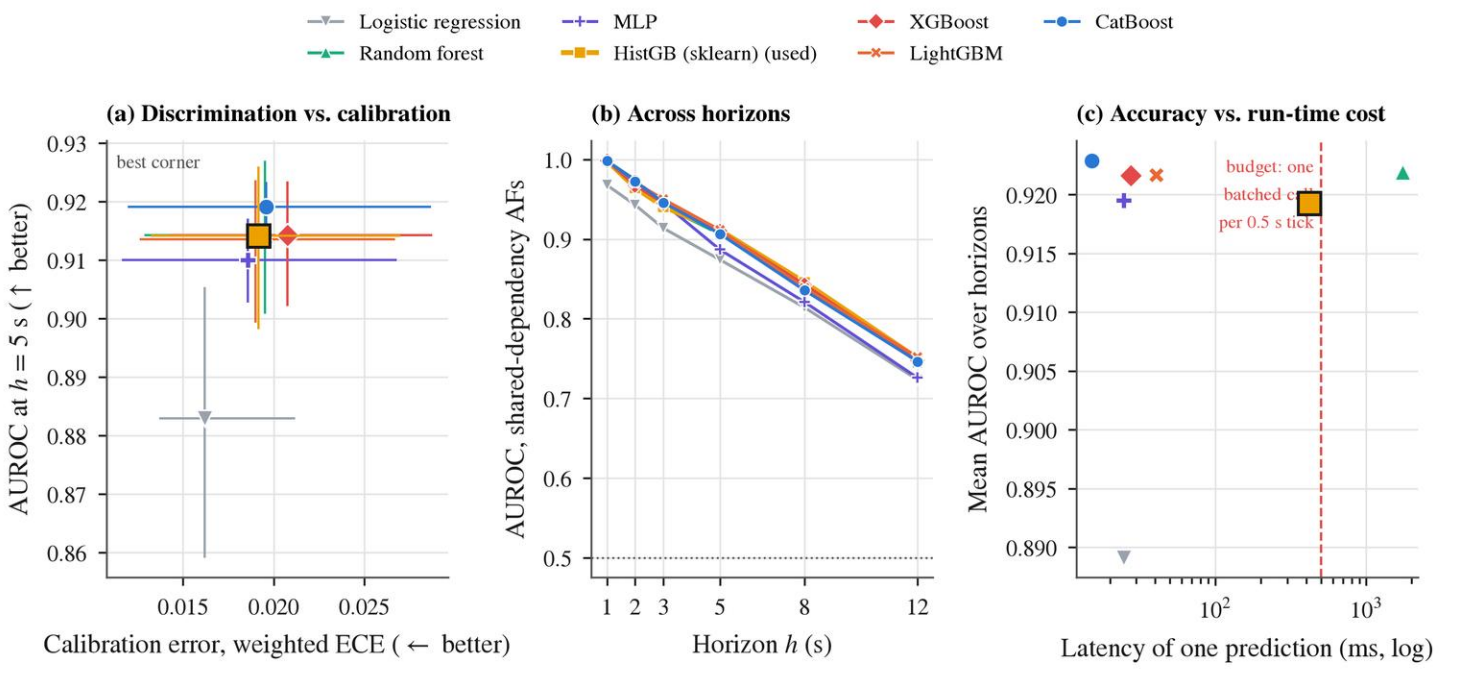

Model A: per-AF features (per-AF predictor)
Model B: + shared-service features (DTIP)
(a) AUROC, all AFs
(b) AUROC, AFs with a shared dependency
(c) ROC at h = 5 s, shared AFs
AUROC (test runs)
1.0
0.9
0.8
0.7
0.6
0.5
chance
1 2 3 5 8 12
Horizon h (s)
True-positive rate
0.0 0.2 0.4 0.6 0.8 1.0
False-positive rate
0.00 0.25 0.50 0.75 1.00
Model A (AUC 0.65)
Model B (AUC 0.91)



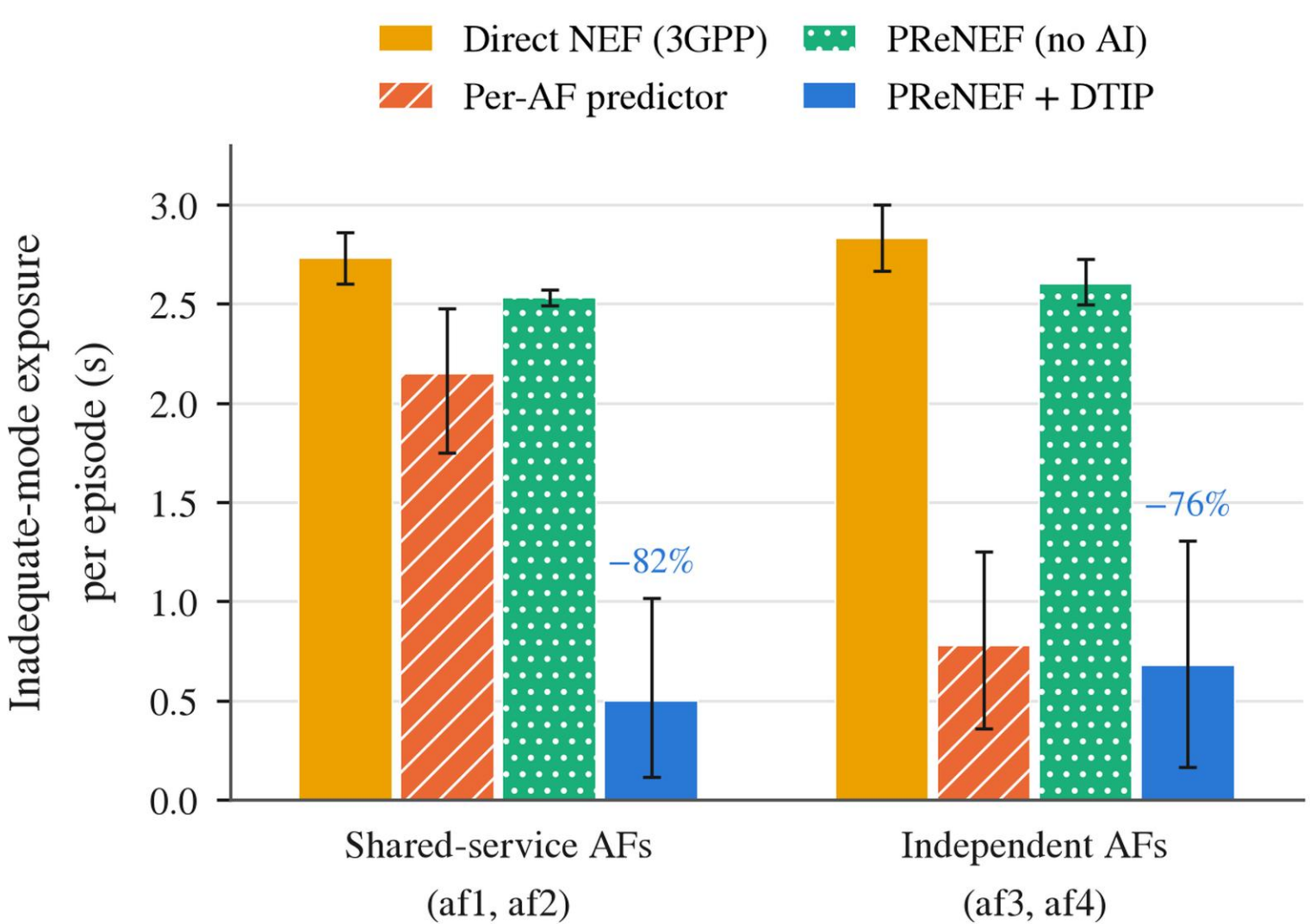
Points on the diagonal = perfectly calibrated; marker size = number of test samples in the bin (bins with < 10 samples hidden).

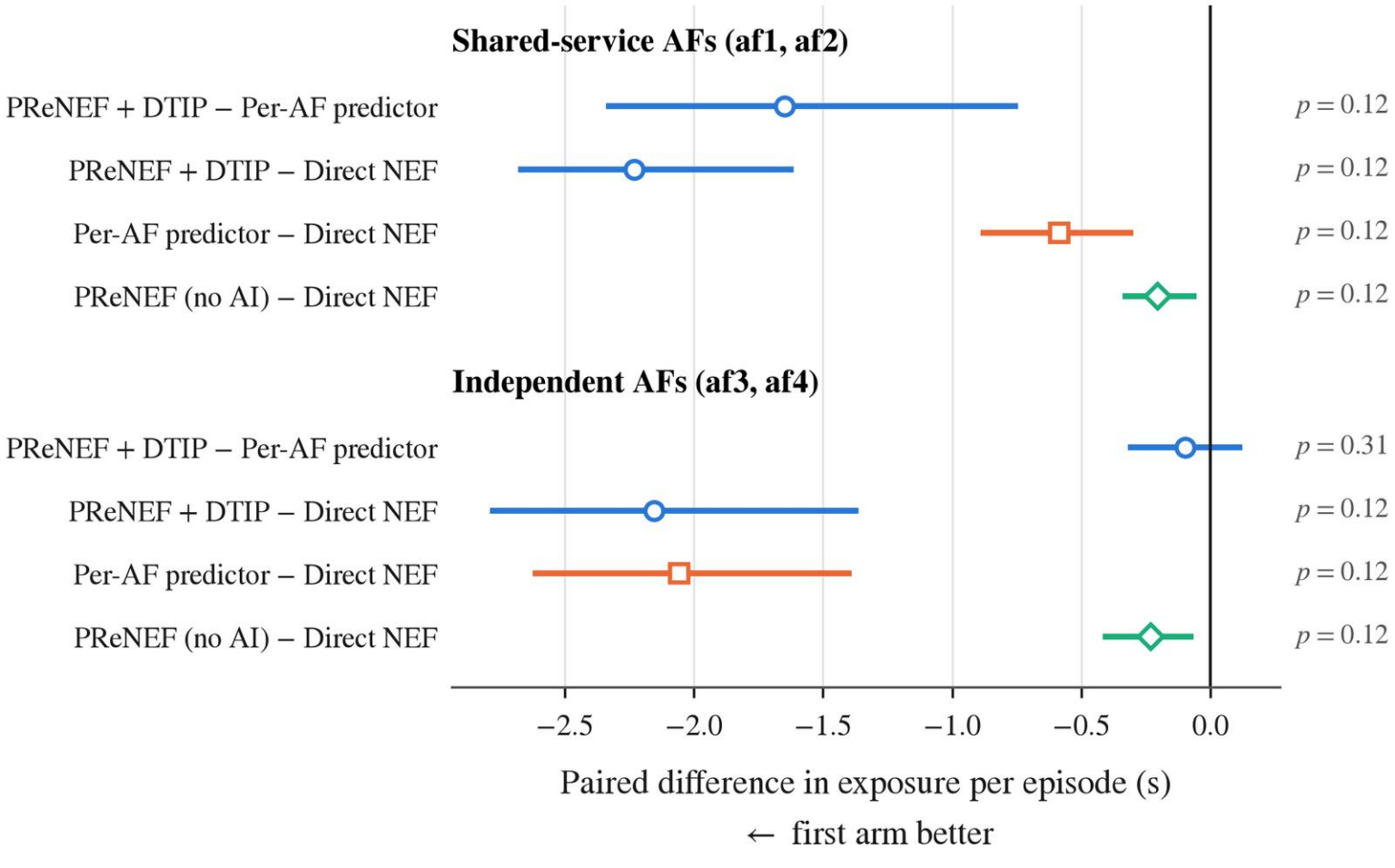

(a) Safe-time prediction (target coverage $\alpha = 0.9$)
Model A: coverage 0.96
Model B: coverage 0.94
safe if T_{safe} below this line
Predicted conservative T_{safe} (s)
0
2
4
6
8
10
12
0–1
1–2
2–3
3–5
5–8
8–12
True time to inadequacy (s)
(b) Uncertainty gate (eq. 19): trust vs. coverage
Coverage of T_{safe}
0.89
0.90
0.91
0.92
0.93
0.94
0.95
0.96
0.97
target 0.9
u_{max}=1.00
u_{max}=1.00
Model A
Model B
0.0
0.2
0.4
0.6
0.8
1.0
Fraction of predictions kept (u ≤ threshold)

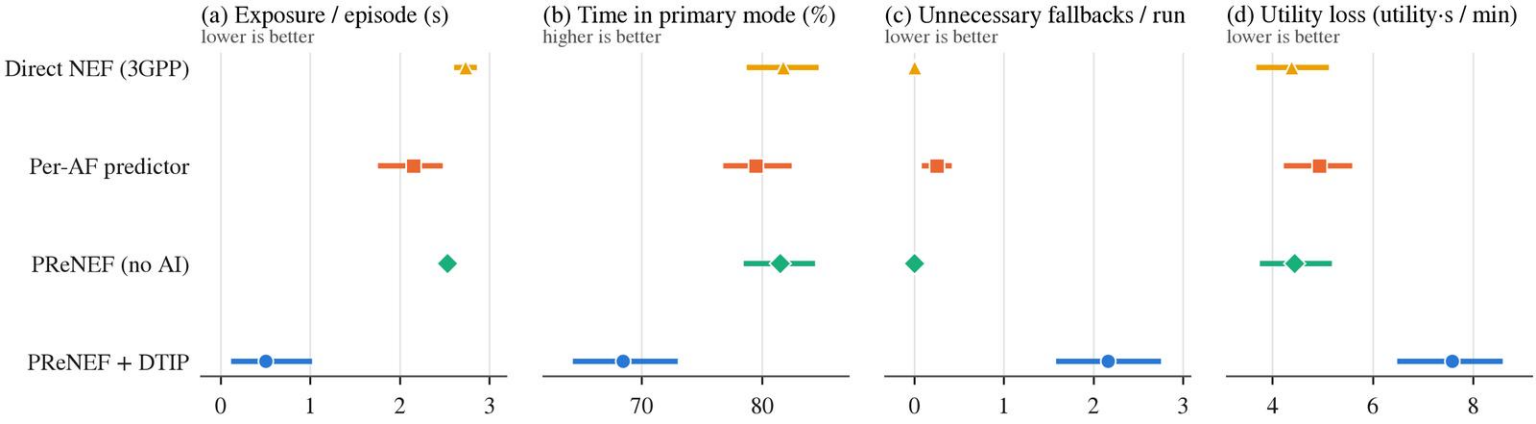

dependency latency
latency slope
latency max (window)
AFs sharing dependency
dependency latency max
dependency down
latency (last)
dependency latency slope
observations (window)
observation age
throughput slope
throughput (last)
loss (last)
shared-service (dependency) feature
per-AF feature
0.00
0.02
0.04
0.06
0.08
AUROC drop when the feature is shuffled ($h = 5$ s)

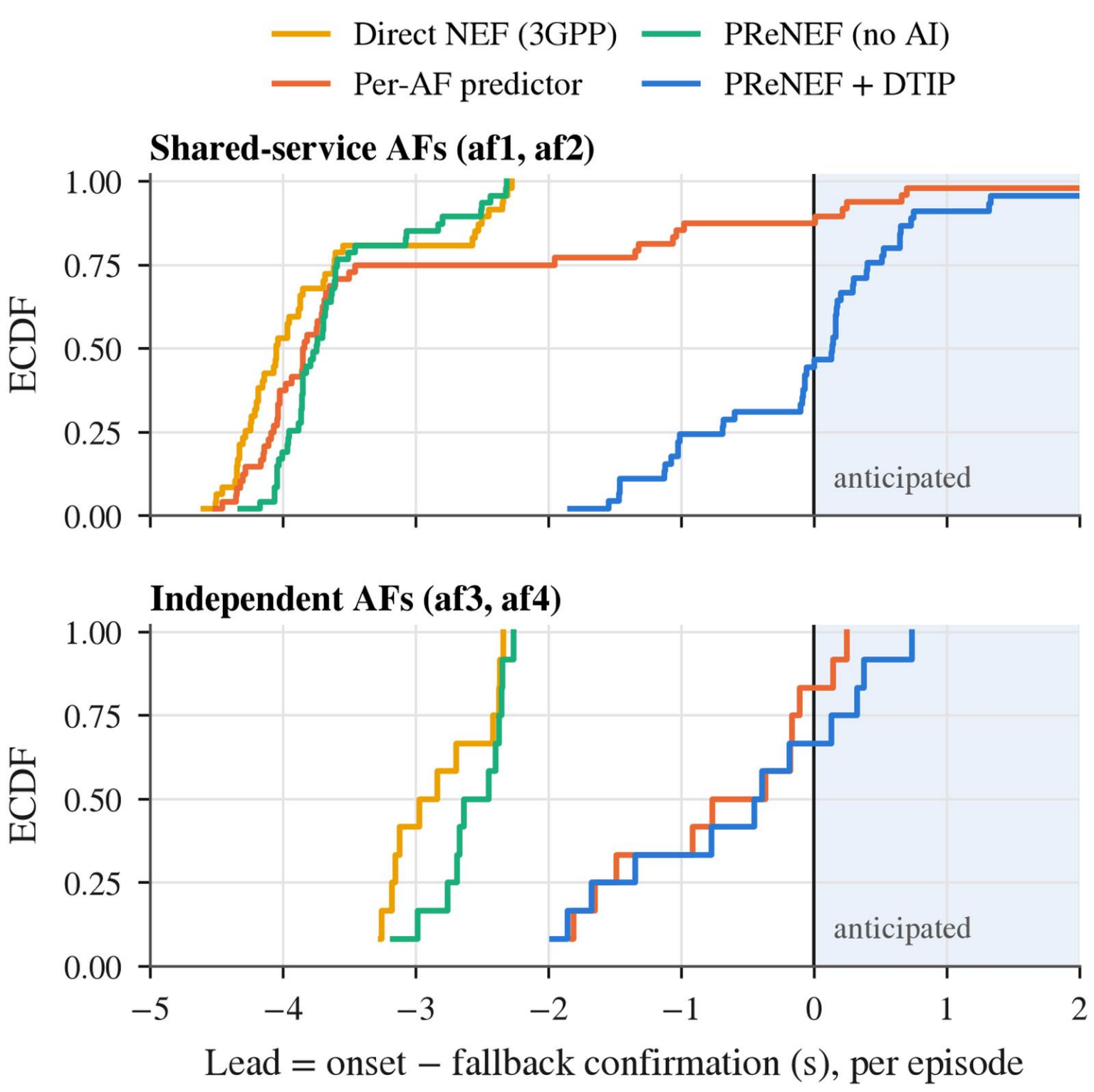


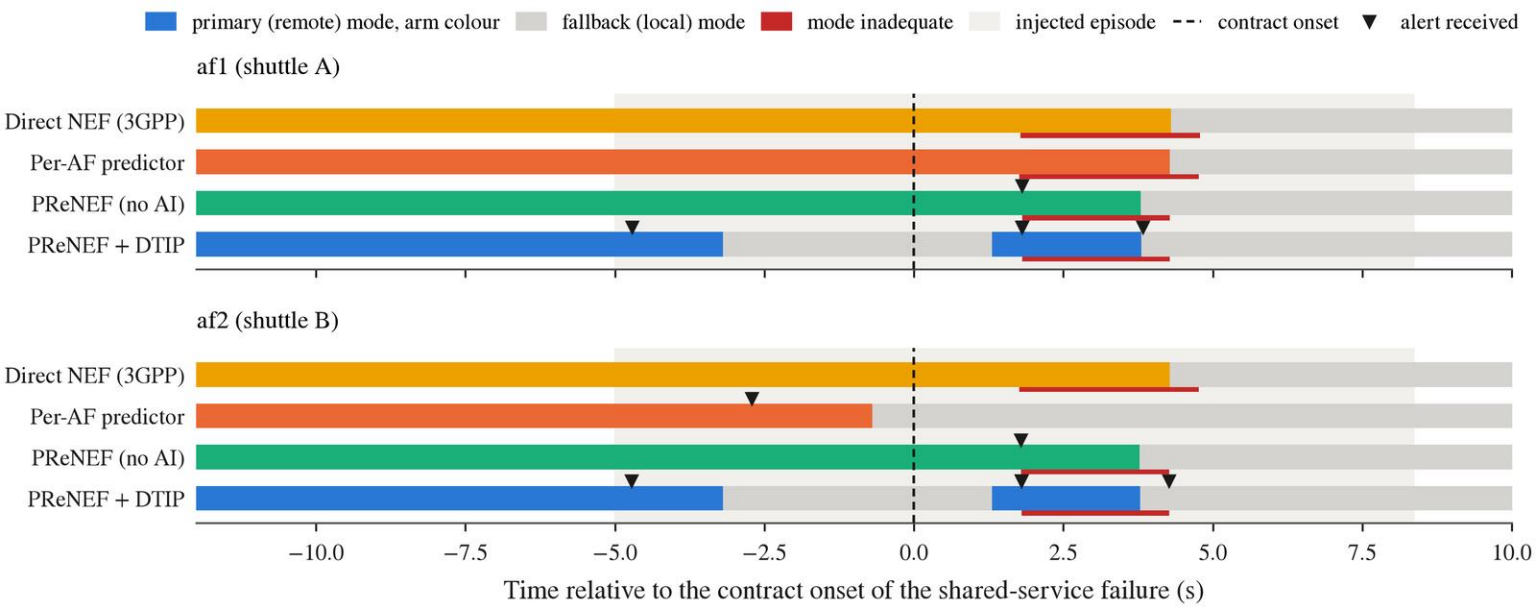
Same data, split, calibration and conformal pipeline for every learner; only the base classifier changes. Error bars: 95 % CI, bootstrap over test runs. Outlined marker = learner used in PReNEF.

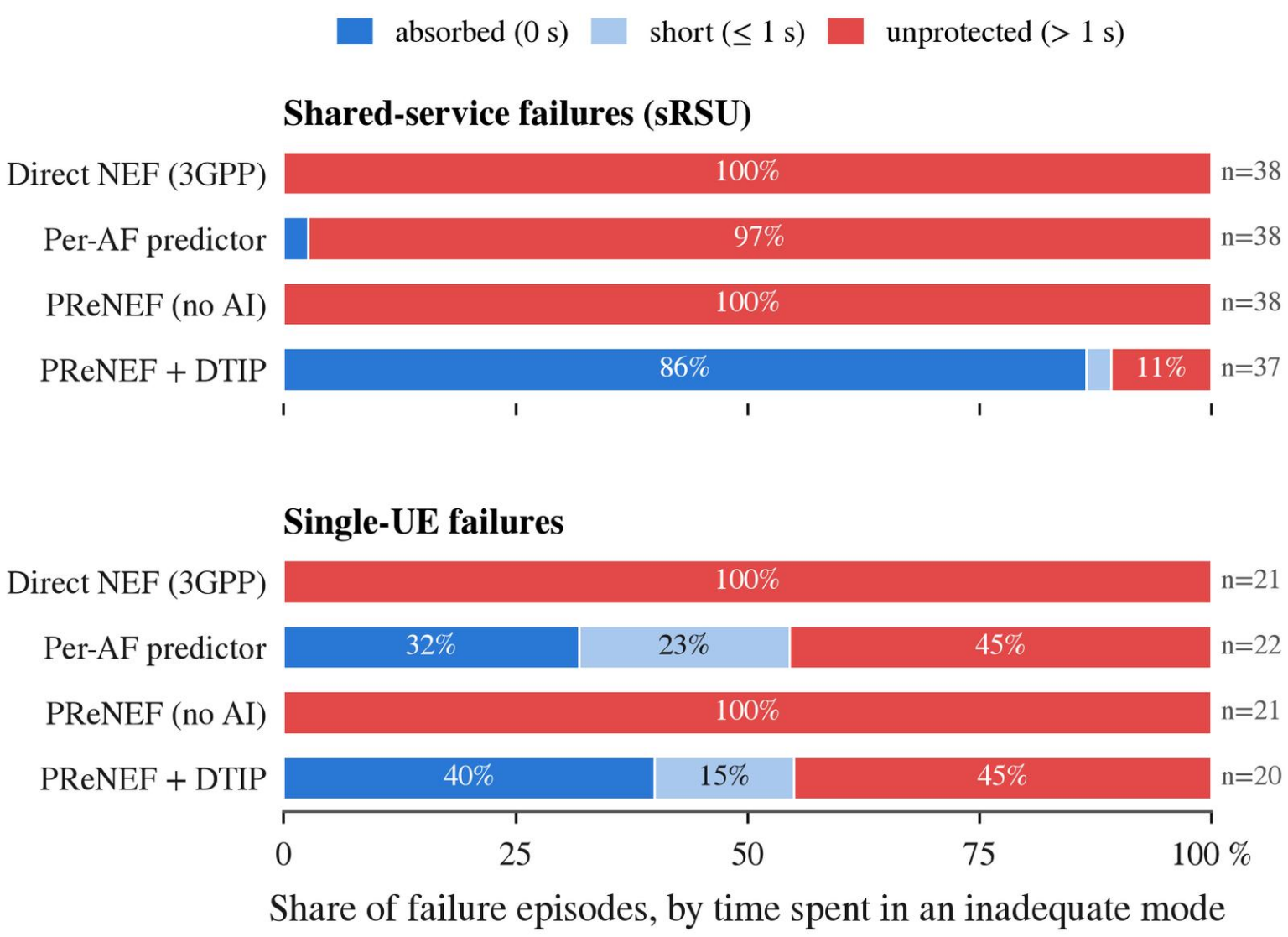

Direct NEF (3GPP)
PReNEF (no AI)
Per-AF predictor
PReNEF + DTIP
Inadequate-mode exposure per episode (s)
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−82%
−76%
Shared-service AFs
(af1, af2)
Independent AFs
(af3, af4)

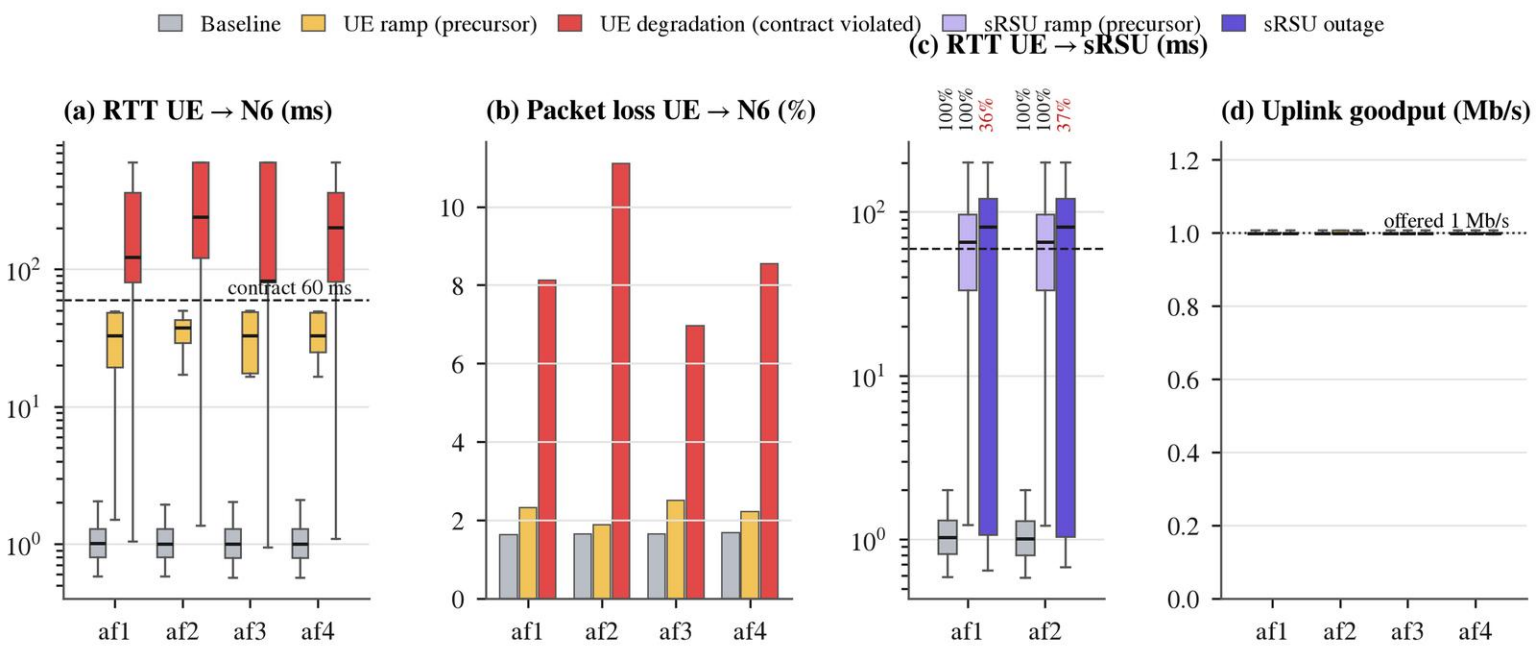

Shared-service AFs (af1, af2)
PReNEF + DTIP − Per-AF predictor
PReNEF + DTIP − Direct NEF
Per-AF predictor − Direct NEF
PReNEF (no AI) − Direct NEF
p = 0.12
p = 0.12
p = 0.12
p = 0.12
Independent AFs (af3, af4)
PReNEF + DTIP − Per-AF predictor
PReNEF + DTIP − Direct NEF
Per-AF predictor − Direct NEF
PReNEF (no AI) − Direct NEF
p = 0.31
p = 0.12
p = 0.12
p = 0.12
−2.5
−2.0
−1.5
−1.0
−0.5
0.0
Paired difference in exposure per episode (s)
← first arm better


(a) Exposure / episode (s)
lower is better
(b) Time in primary mode (%)
higher is better
(c) Unnecessary fallbacks / run
lower is better
(d) Utility loss (utility·s / min)
lower is better
Direct NEF (3GPP)
Per-AF predictor
PReNEF (no AI)
PReNEF + DTIP
0
1
2
3
70
80
0
1
2
3
4
6
8


Direct NEF (3GPP)
PReNEF (no AI)
Per-AF predictor
PReNEF + DTIP
Shared-service AFs (af1, af2)
ECDF
1.00
0.75
0.50
0.25
0.00
anticipated
Independent AFs (af3, af4)
ECDF
1.00
0.75
0.50
0.25
0.00
anticipated
−5
−4
−3
−2
−1
0
1
2
Lead = onset − fallback confirmation (s), per episode

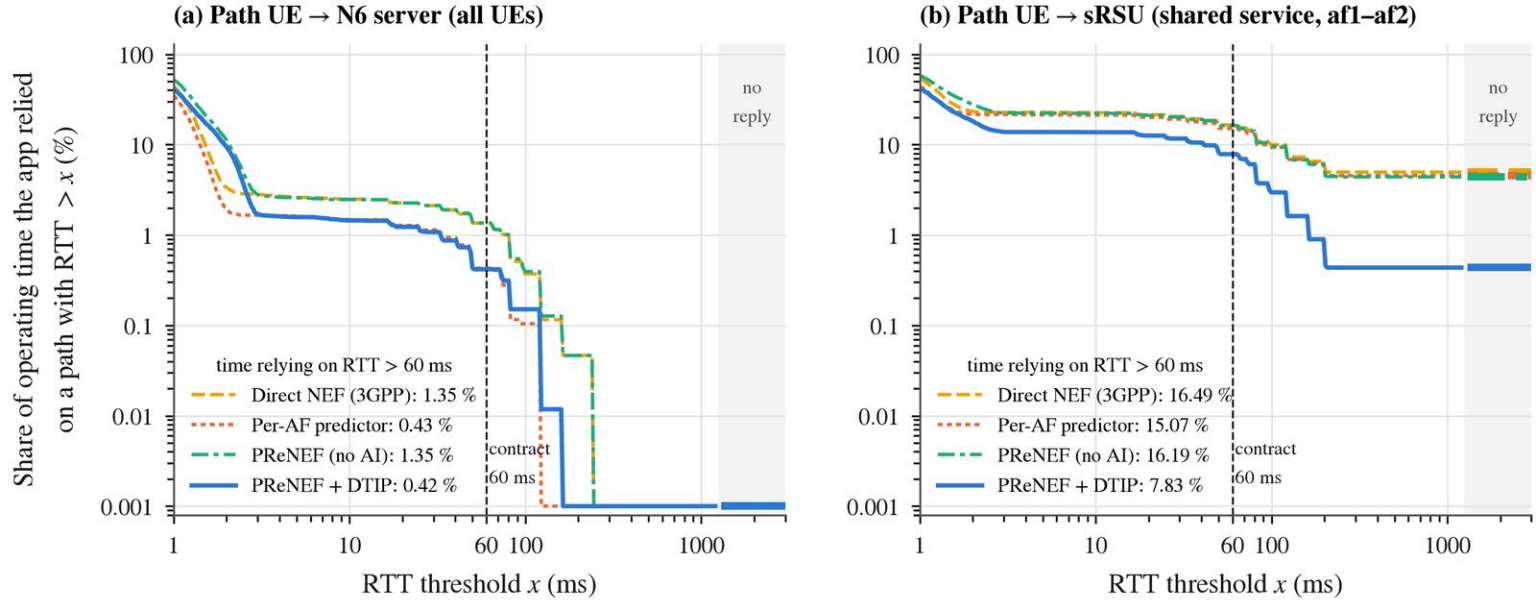

primary (remote) mode, arm colour
fallback (local) mode
mode inadequate
injected episode
contract onset
alert received
af1 (shuttle A)
Direct NEF (3GPP)
Per-AF predictor
PReNEF (no AI)
PReNEF + DTIP
af2 (shuttle B)
Direct NEF (3GPP)
Per-AF predictor
PReNEF (no AI)
PReNEF + DTIP
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
Time relative to the contract onset of the shared-service failure (s)


absorbed (0 s)
short (≤ 1 s)
unprotected (> 1 s)
Shared-service failures (sRSU)
Direct NEF (3GPP)
100%
n=38
Per-AF predictor
97%
n=38
PReNEF (no AI)
100%
n=38
PReNEF + DTIP
86%
11%
n=37
Single-UE failures
Direct NEF (3GPP)
100%
n=21
Per-AF predictor
32%
23%
45%
n=22
PReNEF (no AI)
100%
n=21
PReNEF + DTIP
40%
15%
45%
n=20
0
25
50
75
100 %
Share of failure episodes, by time spent in an inadequate mode

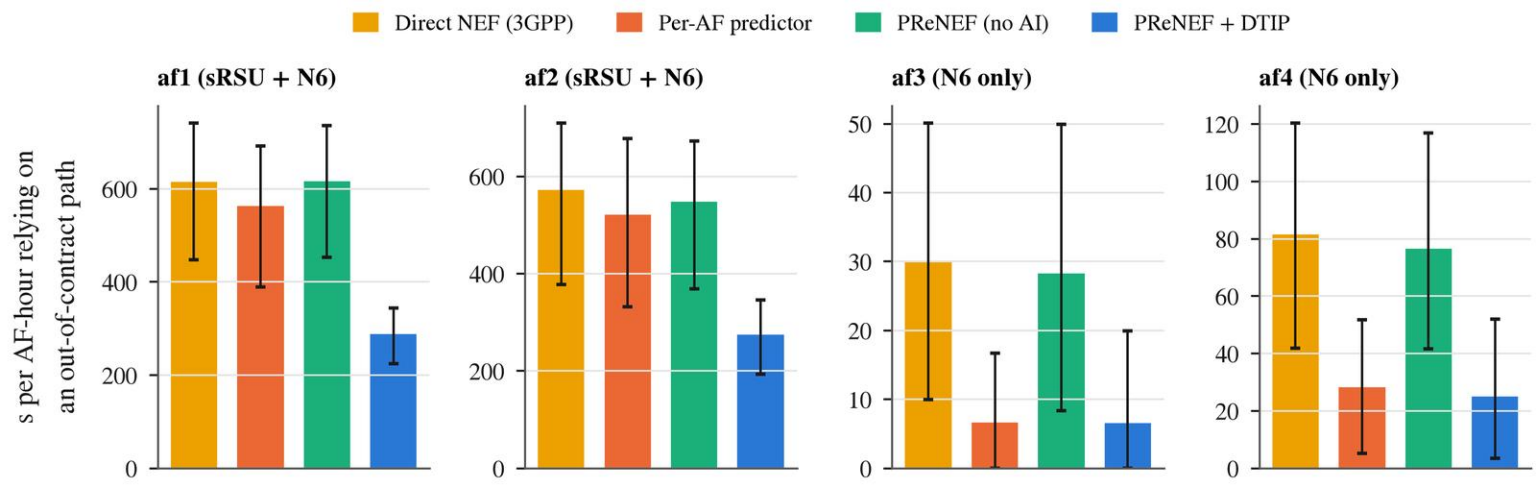


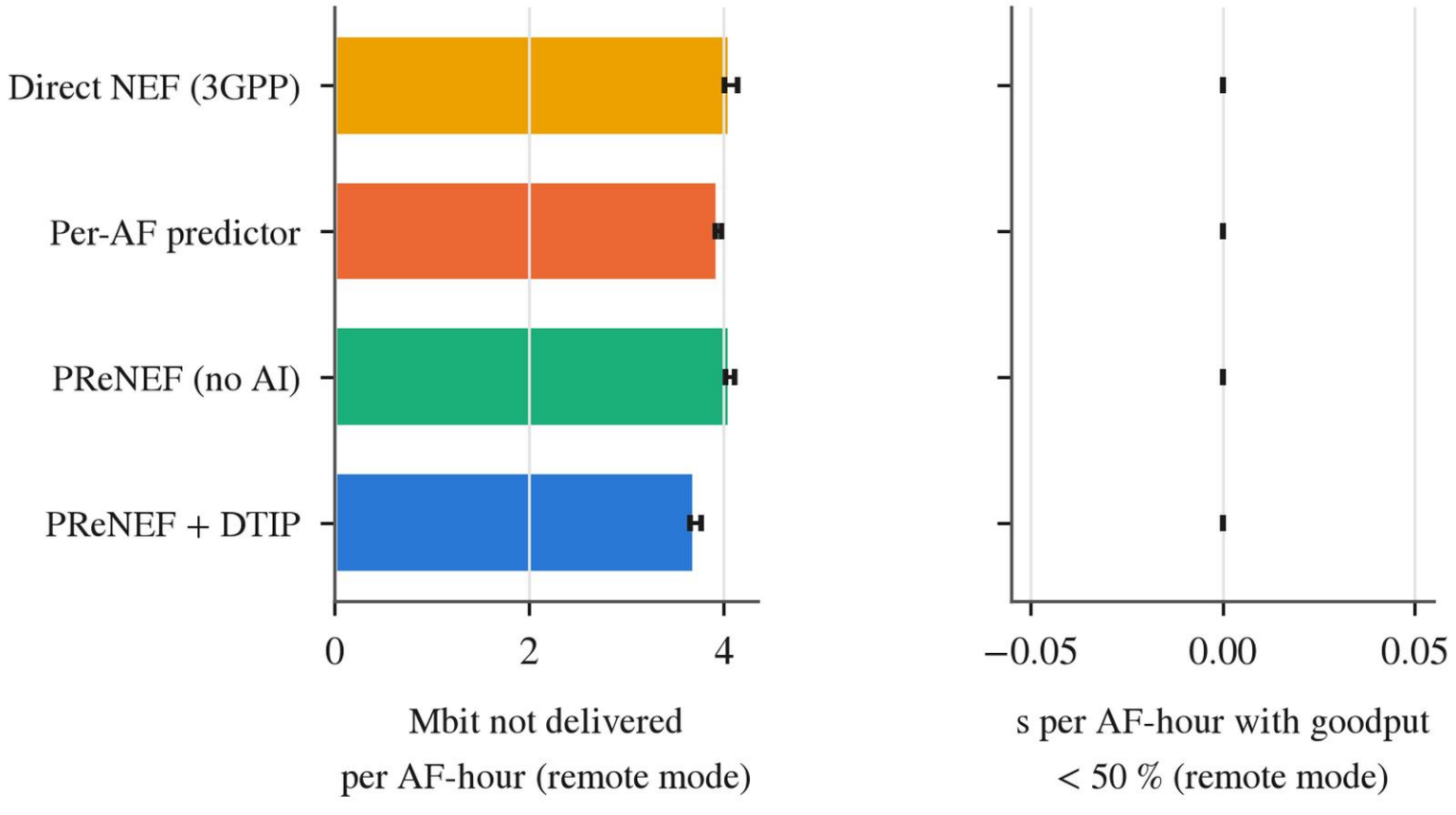
top: % of probes answered

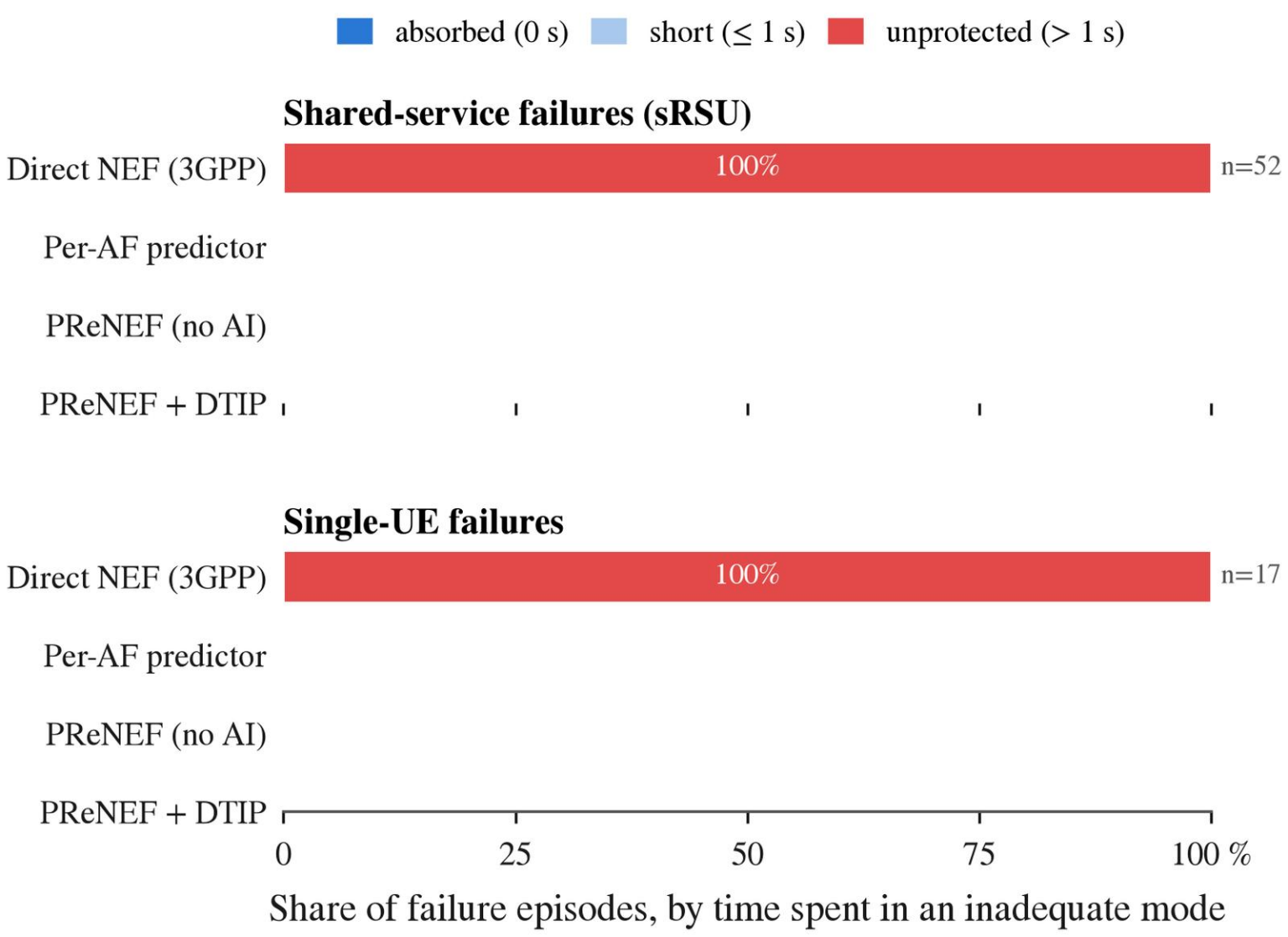
Network conditions per UE and scenario phase, pooled over arms (identical by design). Boxes: median, IQR, whiskers 5–95 %.

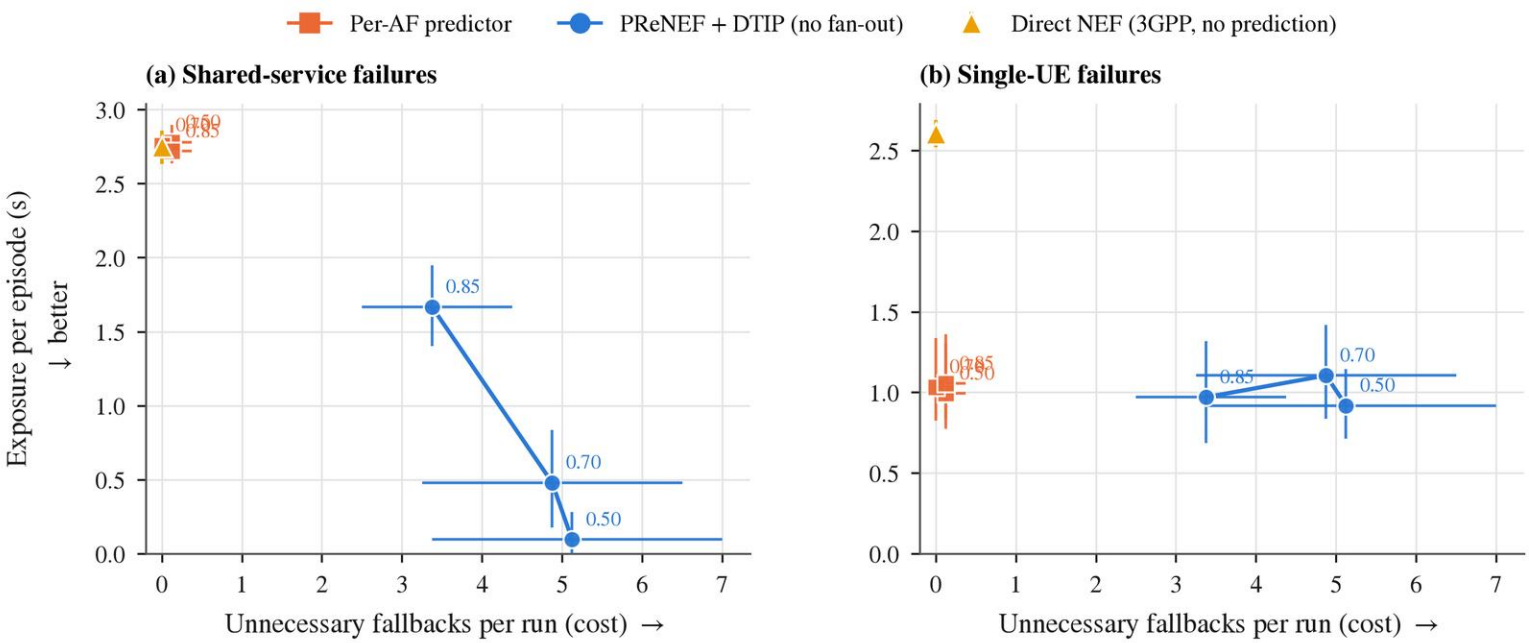


Left end of each curve = share of time in the network-dependent (remote) mode; lower tail = fewer seconds spent depending on an out-of-contract path.


Direct NEF (3GPP)
Per-AF predictor
PReNEF (no AI)
PReNEF + DTIP
af1 (sRSU + N6)
af2 (sRSU + N6)
af3 (N6 only)
af4 (N6 only)
s per AF-hour relying on
an out-of-contract path
0
200
400
600
0
10
20
30
40
50
0
20
40
60
80
100
120


Direct NEF (3GPP)
Per-AF predictor
PReNEF (no AI)
PReNEF + DTIP
0
2
4
Mbit not delivered
per AF-hour (remote mode)
−0.05
0.00
0.05
s per AF-hour with goodput
< 50 % (remote mode)


absorbed (0 s)
short (≤ 1 s)
unprotected (> 1 s)
Shared-service failures (sRSU)
Direct NEF (3GPP)
100%
n=52
Per-AF predictor
PReNEF (no AI)
PReNEF + DTIP
Single-UE failures
Direct NEF (3GPP)
100%
n=17
Per-AF predictor
PReNEF (no AI)
PReNEF + DTIP
0
25
50
75
100 %
Share of failure episodes, by time spent in an inadequate mode


Per-AF predictor
PReNEF + DTIP (no fan-out)
Direct NEF (3GPP, no prediction)
(a) Shared-service failures
(b) Single-UE failures
Exposure per episode (s)
↓ better
Unnecessary fallbacks per run (cost) →
Unnecessary fallbacks per run (cost) →
0.85
0.70
0.50
0.85
0.70
0.50
Numbers next to points: alert threshold p_{alert} (higher = fewer, later alerts)